UTS UNIVERSITY OF TECHNOLOGY SYDNEY

Handbook

Subject

36106 - Machine Learning Algorithms and Applications

8 Credit points

Faculty
TD School

Subject type
Subject

Study level
Postgraduate

Result type
Grade and marks

Total Workload hours
200

Subject description

This subject introduces students to key machine learning algorithms and their application in real-world settings. Participants are guided in developing an intuitive understanding of how the algorithms work, as well as their strengths and weaknesses. In addition to gaining practical experience with the algorithms, students develop an understanding of the basic principles of machine learning and the connections between different algorithms. Additionally, they are exposed to industry standard methodologies for data mining and analytics via readings and assessments. Since data science problems are infused with assumptions, often with ethical and legal implications, due attention is given to questioning the assumptions behind data and approaches used to analyse it.

Availabilities

Click on Select availability in the top right of screen and choose from the location, attendance mode and session options.

Learning outcomes

Upon completion of this subject, graduates will be able to:

1.	Apply an industry standard analytics life cycle methodology for data mining and pattern discovery
2.	Interpret, synthesise and communicate insights extracted from machine learning algorithms in a context-appropriate manner.
3.	Articulate the strengths, weaknesses and assumptions of a selection of machine learning algorithms in relation to structured and unstructured data
4.	Execute and interpret machine learning approaches available for extracting value from data
5.	Demonstrate an appreciation, with examples, of a critical, ethical perspective on decisions made throughout the analytics lifecycle

Learning and teaching activities

Blend of online and face to face activities: This subject is offered through a series of block sessions and blends online with face-to-face learning. Students participate in interactive learning experiences in timetabled on-campus sessions, where they make use of the subject materials that they have already engaged with online. In between campus sessions, students will engage in individual and collaborative online activities designed to support the understanding of the machine learning algorithms and their application in real-world settings
Collaborative work: A strong emphasis is placed on group activities and interaction, given that graduates of this course will need to approach professional projects and challenges from a collaborative and consensus position. Insights obtained and developed within the groups is then reworked by individual students to develop the final summative assessment activity. Group assessments and activities enable students to leverage peer-learning and demonstrate effective skills associated with the topics covered in this subject.
Transdisciplinary approaches: Starting from an elemental perspective on data and data science, students will approach learning from their specific professional and potential future contexts. As the subject progresses, the students will be able to combine their analytical and technical skills in developing and applying various machine-learning algorithms, as well as to consider standards and ethical implications of their work.
An aim of this subject is to help you develop academic and professional language and communication skills in order to succeed at university and in the workplace. To determine your current academic language proficiency, one of the assessment tasks in this subject will be used to assess your level of academic English language. If you receive an unsatisfactory level for English language, you must attend follow up language development activities, in order to pass the subject. These activities are designed to support you to develop your language and communication skills. Students who do not attend 80% of the language development activities will receive a Fail X grade for the subject.

Assessment information

Click on Select availability in the top right of screen and choose from the location, attendance mode and session options.

Requisites

Requisite detail

Before you enrol in this subject, you must ensure you have met its requisites. The requisites for this subject can be viewed on its access conditions page.

To learn more about requisites, please visit the How to use the Handbook guide.

Additional subject details

Attendance requirement

block mode with weekly online work, 1 Wednesday evening and 4 Saturday sessions online or on campus

Apply now **Important dates** **Fee calculator**

UTS acknowledges the Gadigal people of the Eora Nation, the Boorooberongal people of the Dharug Nation, the Bidiagal people and the Gamaygal people, upon whose ancestral lands our university stands. We would also like to pay respect to the Elders both past and present, acknowledging them as the traditional custodians of knowledge for these lands.

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